3.2 Characteristics of Functions

Name _____ Period ____ Date _____

Algebra I 3.2 · Student sheet

I can…

- find the x- and y-intercepts from a graph.
- tell where a function is positive or negative and increasing or decreasing.
- describe the end behavior of a function.

Vocabulary · fill in the missing word

x-intercept — where the graph _____ the x-axis (y = 0)

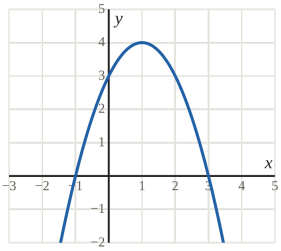
y-intercept — where the graph _____ the y-axis (x = 0)

Positive / negative — the graph is above / below the _____

Increasing / decreasing — the _____ goes up / down as you move to the right

End behavior — what y does as x goes far to the left and far to the _____

Part 1 · Graph story



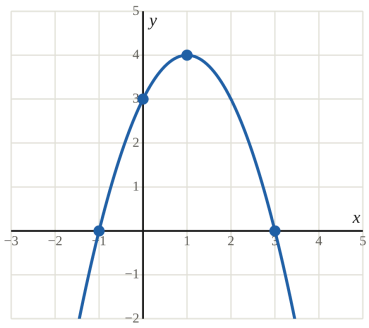
Feature	Answer
x-intercepts	
y-intercept	
Positive / negative	
Increasing / decreasing	
End behavior	

A story this graph could tell: _____

Part 2 · Notes and examples (fill in each note, then work the example)

Example 1

Find the x- and y-intercepts of the graph of $f(x) = -x^2 + 2x + 3$.



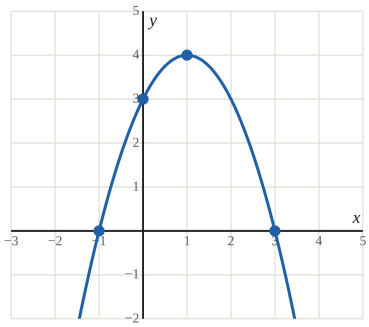
Positive and negative: A function is positive where its graph is _____ the x-axis.

Increasing and decreasing: Increasing: the graph goes _____ as you move to the right.

Intervals: Describe intervals with _____-values.

Example 2

Use the graph from Example 1. Where is the function positive? Negative? Increasing? Decreasing? Describe its end behavior.



Example 3

Describe $f(x) = -x + 3$: intercepts, where it is positive and negative, whether it increases or decreases, and its end behavior.

Part 3 · Partner practice

Partner 1

Find the intercepts of $y = 3x + 6$.

Partner 2

For $f(x) = |x| - 1$, where is f negative? Where is it decreasing?